EMMANUEL A. DE BELEN

Electronics Engineering Graduate | Aspiring Software Engineer
emmldbln.work@gmail.com | 63 966 698 8831 | Taytay, Rizal

PROFESSIONAL SUMMARY

Electronics Engineering graduate with hands-on experience in full-stack web development, IoT system development, software development, broadcast engineering, and technical troubleshooting. Experienced in developing applications using PHP, JavaScript, TypeScript, React, Next.js, MySQL, Firebase, and REST APIs, with practical experience in database-driven systems and cloud deployment. Research presented and accepted at IEEE International Geoscience and Remote Sensing Symposium 2026.

EDUCATION

Bachelor of Science in Electronics Engineering Aug 2019 – May 2025
Adamson University – Ermita, Manila

TECHNICAL SKILLS

IT Support & Troubleshooting: Windows | Hardware/Software Troubleshooting | Cisco Networking (Familiarity)
Programming: Java | Python | C++ | PHP | JavaScript | TypeScript | SQL | React | Next.js | Arduino | AI-Assisted
Web & Cloud Technologies: HTML | CSS | REST APIs | Google APIs | Vercel | GitHub
Databases: MySQL | PostgreSQL | Firestore
Engineering Tools: LabVIEW | AutoCAD | Microsoft Office | QGIS | NetBeans | MATLAB | Visual Studio Code | XAMPP

PROJECT EXPERIENCE

ThermaMist: Misting System with Ground Sensor Terminals for Localized Intra-Urban Heat Assessment and Cool Edge Effect (CAPSTONE)

- Developed an IoT-based environmental monitoring and automated misting system using ESP32, DHT22, solar irradiance sensor, and anemometer.
- Implemented sensor data acquisition and automated misting control based on Heat Index and Wet Bulb Globe Temperature environmental conditions.
- Integrated Google Sheets and Apps Script for timestamped environmental data logging and monitoring.
- Used QGIS to organize and visualize geospatial environmental data for analysis.
  Contributed to the research paper, which was accepted and presented at IEEE IGARSS 2026.

Evergreen: Full-Stack Web Application (Personal)

- Developed a production-ready full-stack web application using Next.js, React, TypeScript, and Firebase.
- Integrated Google OAuth 2.0 and Google Drive API for secure cloud-based media synchronization and management.
- Implemented REST API routes, Firestore data management, responsive UI components, and Vercel deployment.

Marcid Blue: Water Station Management System

- Developed a PHP, MySQL, and JavaScript-based business management system for daily water station operations.
- Designed and implemented modules for daily sales, customer records, deliveries, payments, expenses, debts, driver remittances, and daily closing.
- Developed MySQL database structures and backend logic for recording transactions, tracking customer balances, and calculating daily financial summaries.
- Implemented admin authentication, session management, role-based access controls, customer autocomplete, transaction validation, and data filtering.
- Built responsive business interfaces using HTML, CSS, and JavaScript, with features for daily record management and printable financial reports.

WORK EXPERIENCE

Broadcast Engineering Intern — ZOE Broadcasting Network June 2024 – July 2024

- Supported live broadcast in the TOC (Technical Operations Center)
- Monitored transmitters and broadcast systems (Includes Troubleshooting and Maintenance).
- Collaborated on technical reports, presentations, and feasibility studies.

SEMINAR AND AWARDS

- Consistent Honor Student (SHS & HS)
- Best Speaker in Debate
- Broadcast And Acoustics 2024 Seminar
- Above and Beyond: Discovering Next-Generation Electronics Profession 2024 Seminar
- Smart Technology, Telecommunications 2024 Seminar
- IoT Seminars